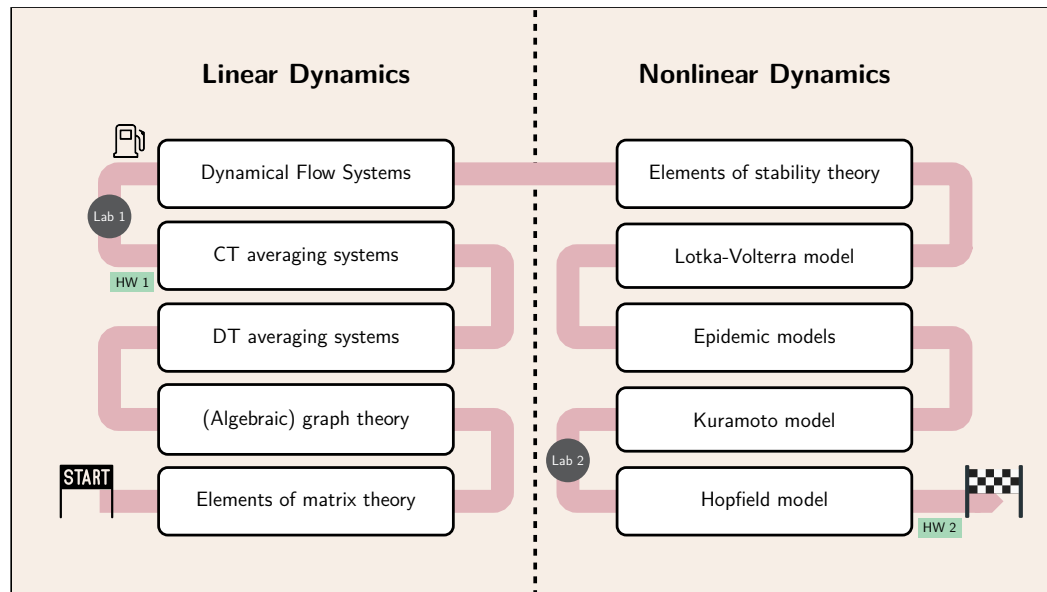
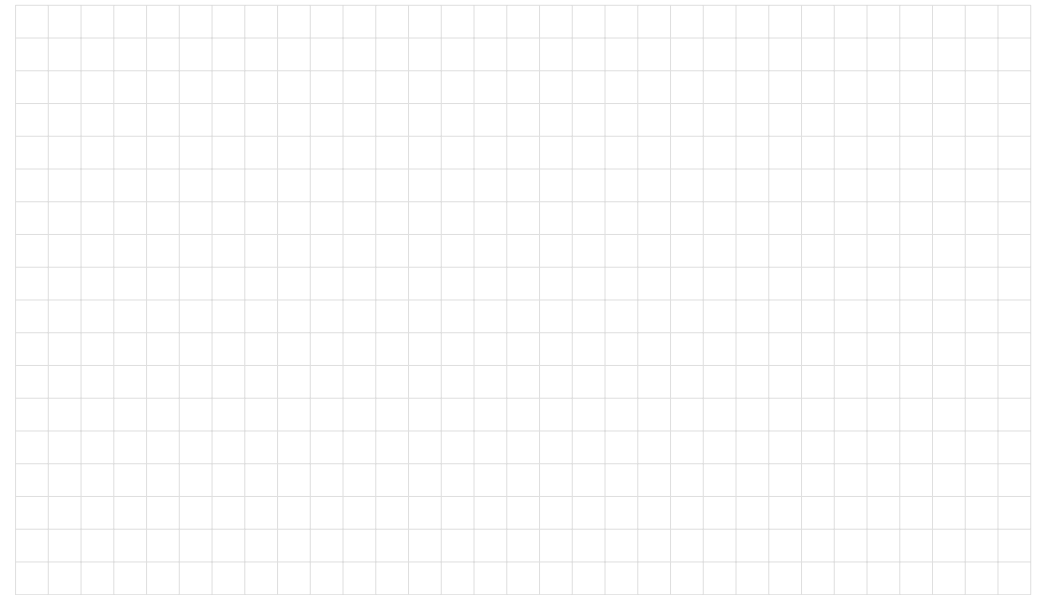


Elements of Algebraic Graph Theory

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Network Dynamical Systems
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Covered topics

- ▷ DT averaging systems: recap and examples
- ▷ Preliminaries on substochastic matrices
- ▷ Systems achieving asymptotic consensus
- ▷ Systems reaching asymptotic disagreement



CT averaging systems: intro

- ▷ $A \geq 0$ adjacency matrix of weighted digraph $G = (V, E, \{a_{ij}\})$

Self loop of G don't matter for CT avg sys $\leftarrow$

$$\dot{x}_i(t) = \sum_{j=1}^n a_{ij}(x_j(t) - x_i(t)), \quad i = 1, \dots, n, \quad t \in \mathbb{R}, t \geq 0$$

$$\Rightarrow \dot{x}(t) = (A - \text{diag}(A\mathbf{1}_n))x(t), \quad t \in \mathbb{R}, t \geq 0, \quad x(0) = x_0 \in \mathbb{R}^n$$

- ▷ $x^* = \alpha \mathbf{1}_n, \alpha \neq 0$ equilibrium point of the system $\rightarrow (A - \text{diag}(A\mathbf{1}_n))\alpha \mathbf{1}_n = 0$
- ▷ convergence to $x^* \Rightarrow$ asymptotic consensus $= \alpha A\mathbf{1}_n - \alpha \text{diag}(A\mathbf{1}_n)\mathbf{1}_n = 0$

When do we achieve asymptotic consensus? $\forall v \in \mathbb{R}^n: \text{diag}(v)\mathbf{1}_n = v$

From DT to CT averaging systems

Consider a DT avg system: $x(k+1) = Ax(k) \quad k=0,1,\dots$ A row-stoch

Consider a more general DT avg sys. in which the time steps are $0, \Delta t, 2\Delta t, 3\Delta t, \dots$

where $\Delta t \in \mathbb{R}_+$ $x(t+\Delta t) = Ax(t) \quad t = 0, \Delta t, 2\Delta t, \dots$

Now we compute: $\frac{x(t+\Delta t) - x(t)}{\Delta t} = \frac{Ax(t) - x(t)}{\Delta t} = \frac{1}{\Delta t} (A - I)x(t)$

We assume that $A = \Delta t \bar{A}$ where $\bar{A}$ is indep. of $\Delta t \Rightarrow \frac{1}{\Delta t} (A - I) = \frac{1}{\Delta t} (\Delta t \bar{A} - I) = \bar{A} - \text{diag}(\bar{A}\mathbf{1}_n)$

$A\mathbf{1}_n = \mathbf{1}_n$
 $\Delta t \bar{A}\mathbf{1}_n = \mathbf{1}_n$
 $\text{diag}(\Delta t \bar{A}\mathbf{1}_n) = \text{diag}(\mathbf{1}_n) = I$

By plugging prev. identities in ⑤ $\frac{x(t+\Delta t) - x(t)}{\Delta t} = (\bar{A} - \text{diag}(\bar{A}\mathbf{1}_n))x(t)$

For $\Delta t \rightarrow 0^+$ $\dot{x}(t) = (\bar{A} - \text{diag}(\bar{A}\mathbf{1}_n))x(t) \leadsto$ CT avg. sys

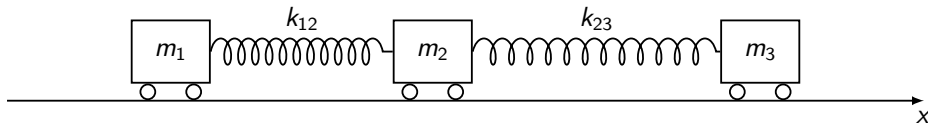
Component-wise $\dot{x}_i(t) = [(\bar{A} - \text{diag}(\bar{A}\mathbf{1}_n))x(t)]_i$

$$= [\bar{A}x(t)]_i - [\text{diag}(\bar{A}\mathbf{1}_n)x(t)]_i$$

$$= \sum_{j=1}^n \bar{a}_{ij} x_j(t) - \sum_{j=1}^n \bar{a}_{ij} x_i(t)$$

$$= \sum_{j=1}^n \bar{a}_{ij} (x_j(t) - x_i(t)) \quad i = 1, \dots, n$$

CT averaging systems: mechanical networks



- ▷ x_i = position of cart i , m_i = mass of cart i
- ▷ b_i = viscous damping of cart i , k_{ij} = elastic constant of spring between i and j

$$m_i \ddot{x}_i = -b_i \dot{x}_i + \sum_j k_{ij} (x_j - x_i) \xrightarrow{b_i \gg m_i} \dot{x}_i = \sum_j \frac{k_{ij}}{b_i} (x_j - x_i)$$

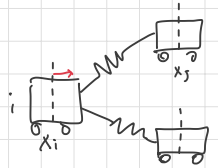
Assumptions: (i) Carts move on an ideal plane (no friction)

(ii) The spring connecting m_i to m_j is at rest when $x_i = x_j$

Let G be the digraph in which the nodes are the carts and there is an edge connecting cart i to cart j if there is a spring connecting i to j



By Newton's law we have that: $m_i \ddot{x}_i = F_{\text{elastic},i} + F_{\text{damping},i}$

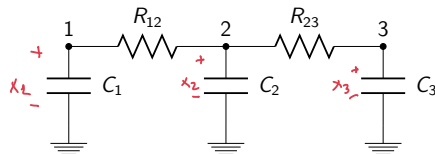


$$\begin{cases} m_1 \ddot{x}_1 = k_{12} (x_2 - x_1) - b_1 \dot{x}_1 \\ m_2 \ddot{x}_2 = k_{21} (x_1 - x_2) + k_{23} (x_3 - x_2) - b_2 \dot{x}_2 \\ m_3 \ddot{x}_3 = k_{32} (x_2 - x_3) - b_3 \dot{x}_3 \end{cases}$$

$$m_i \ddot{x}_i + b_i \dot{x}_i = \sum_{j \in N^{\text{out}}(i)} k_{ij} (x_j - x_i) \rightarrow \frac{m_i}{b_i} \dot{x}_i + x_i = \sum_{j \in N^{\text{out}}(i)} \frac{k_{ij}}{b_i} (x_j - x_i) \xrightarrow{\text{if } b_i \gg m_i} \dot{x}_i = \sum_{j \in N^{\text{out}}(i)} \frac{k_{ij}}{b_i} (x_j - x_i) \quad \text{CT AVG. SYS}$$

$$\dot{X} = (A - \text{diag}(A \mathbf{1} \mathbf{1}^T)) \quad A = \begin{bmatrix} 0 & \sim & 0 \\ \sim & 0 & \sim \\ 0 & \sim & 0 \end{bmatrix}$$

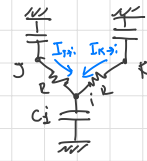
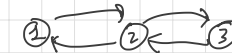
CT averaging systems: electrical networks



- ▷ x_i = electric potential of node i
- ▷ C_i = capacitance of node i , R_{ij} = resistance of branch between i and j

$$C_i \dot{x}_i = \sum_j \frac{1}{R_{ij}} (x_j - x_i) \longrightarrow \dot{x}_i = \sum_j \frac{1}{C_i R_{ij}} (x_j - x_i)$$

Let G be the digraph in which the nodes are the circuit nodes and there is an edge (i,j) if there is a resistor connecting i to j



For a generic Capacitor i : $C_i \dot{x}_i = \sum_{j \in N^{\text{out}}(i)} I_{j \rightarrow i}$

We have $I_{j \rightarrow i} = \frac{1}{R_{ij}} (x_j - x_i)$

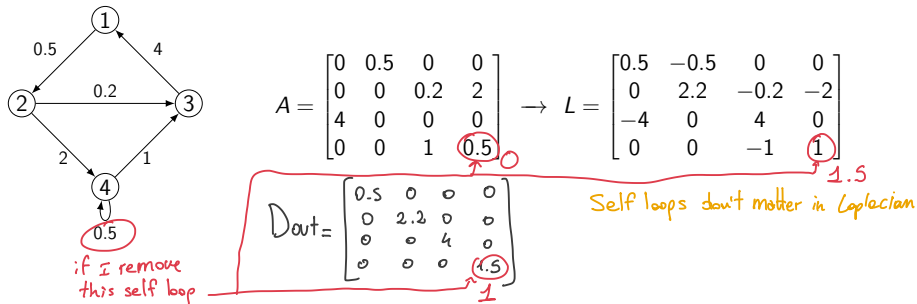
$$\text{Thus: } C_i \dot{x}_i = \sum_{j \in N^{\text{out}}(i)} I_{j \rightarrow i} = \sum_{j \in N^{\text{out}}(i)} \frac{1}{R_{ij}} (x_j - x_i)$$

$$\dot{x}_i = \sum_{j \in N^{\text{out}}(i)} \frac{1}{C_i R_{ij}} (x_j - x_i) \quad \text{CT AVG SYS}$$

Laplacian matrix and its properties

CT averaging system: $\dot{x}(t) = (A - \text{diag}(A\mathbf{1}_n))x(t) = -Lx(t)$

$$L = \text{diag}(A\mathbf{1}_n) - A = D_{\text{out}} - A = \text{Laplacian matrix of weighted digraph } G$$



Laplacian matrix and its properties

Lemma (Properties of Laplacian matrices): Let G be a weighted digraph and let L be the Laplacian matrix of G . The following properties hold:

- (i) $L\mathbf{1}_n = \mathbf{0}_n$;
- (ii) $L^T\mathbf{1}_n = \mathbf{0}_n$ if and only if G is weight-balanced ($D_{\text{out}} = D_{\text{in}}$);
- (iii) if G is undirected ($A = A^T$) then $L = L^T \succeq 0$;
- (iv) $0 \in \text{spec}(L)$ and $\Re(\mu) > 0$ for all $\mu \in \text{spec}(L) \setminus \{0\}$;
- (v) 0 is a semisimple eigenvalue of L ;
- (vi) 0 is a simple eigenvalue of L if and only if G contains a globally reachable node.

▷ **Note:** A Laplacian matrix can be defined also without a digraph G . It is a matrix $L = [\ell_{ij}] \in \mathbb{R}^{n \times n}$ satisfying (1) $L\mathbf{1}_n = \mathbf{0}_n$, (2) $\ell_{ii} \geq 0$ for all i , (3) $\ell_{ij} \leq 0$ for all $i \neq j$

Proof:

$$(i) L \mathbf{1}_n = (\text{diag}(A \mathbf{1}_n)) \mathbf{1}_n = (\text{diag}(A \mathbf{1}_n)) \mathbf{1}_n - A \mathbf{1}_n = \mathbf{0}_n$$

$$(ii) L^T \mathbf{1}_n = (\text{diag}(A \mathbf{1}_n) - A^T) \mathbf{1}_n = \underbrace{A \mathbf{1}_n}_{\substack{\text{dout}(1) \\ \vdots \\ \text{dout}(n)}} - \underbrace{A^T \mathbf{1}_n}_{\substack{\text{din}(1) \\ \vdots \\ \text{din}(n)}} = \mathbf{0}_n \iff \text{dout}(i) = \text{din}(i) \quad \forall i \in V$$

(iii) Assume $A = A^T$

$$L^T = \text{diag}(A \mathbf{1}_n) - A^T = L$$

$$\text{Consider any } x \in \mathbb{R}^n \text{ and note that: } [Lx]_i = [(\text{diag}(A \mathbf{1}_n) - A)x]_i = [\text{diag}(A \mathbf{1}_n)x]_i - [Ax]_i = \sum_{j=1}^n a_{ij} x_j - \sum_{j=1}^n a_{ij} x_j = \sum_{j=2}^n a_{ij} (x_i - x_j)$$

$$\text{Now consider: } x^T Lx = \sum_{i=1}^n x_i [Lx]_i = \sum_{i=1}^n x_i \left(\sum_{j=2}^n a_{ij} (x_i - x_j) \right) = \sum_{i=1}^n x_i \sum_{j=2}^n a_{ij} x_i - \sum_{i=1}^n x_i \sum_{j=2}^n a_{ij} x_j$$

$$\begin{aligned} &= \sum_{i=1}^n \sum_{j=2}^n a_{ij} x_i^2 - \sum_{i=1}^n \sum_{j=2}^n a_{ij} x_i x_j \\ &\quad \left(\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i^2 + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i \right) \xrightarrow{(a_{ij} = a_{ji})} \frac{1}{2} \sum_{i=1}^n \sum_{j=2}^n a_{ji} x_i^2 \xrightarrow{(i \leftrightarrow j)} \frac{1}{2} \sum_{j=1}^n \sum_{i=2}^n a_{ij} x_j^2 \\ &= \frac{1}{2} \sum_{i,j=1}^n a_{ij} x_i^2 + \frac{1}{2} \sum_{i,j} a_{ij} x_j^2 - \sum_{i,j=2}^n a_{ij} x_i x_j \\ &= \frac{1}{2} \sum_{i,j=1}^n a_{ij} (x_i - x_j)^2 \geq 0 \end{aligned}$$

(iv) Since $L \mathbf{1}_n = \mathbf{0}_n = 0 \cdot \mathbf{1}_n \Rightarrow \mathbf{1}_n$ is an eigenvector of L with eigenvalue 0

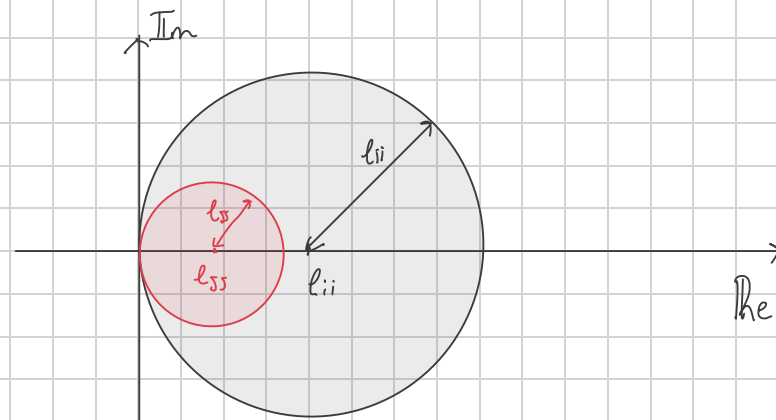
$$\Rightarrow 0 \in \text{spec}(L)$$

disk centered at l_{ii} with radius $R_i = \sum_{j \neq i} |l_{ij}|$

By Gershgorin disk thm: $\text{spec}(L) \subseteq \bigcup_{i=1}^n D_i$, $D_i = \left\{ z \in \mathbb{C} : |z - l_{ii}| \leq \sum_{j \neq i} |l_{ij}| \right\}$

$$\text{Note that: } l_{ii} = \sum_{j=1}^n a_{ij} - a_{ii} = \sum_{j \neq i} a_{ij}$$

$$R_i = \sum_{j \neq i} |l_{ij}| = \sum_{j \neq i} |-a_{ij}| = \sum_{j \neq i} a_{ij}$$



Thus, by the form of D_i , it follows that $\forall \mu \in \text{spec}(L) \setminus \{0\}$ it holds $\text{Re}(\mu) > 0$

(v) Won't be proven

(vi) "G has a globally reachable node $\Rightarrow 0 \in \text{spec}(L)$ simple"

Define $A_L = I - \varepsilon L$ where $0 < \varepsilon < 1/l_{\max}$, $l_{\max} = \max_i l_{ii}$

A_L is row-stochastic, indeed: i) $A_L \mathbf{1}_n = (I - \varepsilon L) \mathbf{1}_n = \mathbf{1}_n - \varepsilon L \mathbf{1}_n = \mathbf{1}_n$

$$\text{ii) } [A_L]_{ij} = \begin{cases} -\varepsilon l_{ij} = \varepsilon a_{ij} > 0 & i \neq j \\ 1 - \varepsilon l_{ii} > 0 & i = j \end{cases}$$

$\Leftrightarrow \varepsilon l_{ii} < 1 \Leftrightarrow \varepsilon < 1/l_{ii}$
 $\varepsilon < 1/l_{\max} \leq 1/l_{ii}$

Let G_{A_L} be the weighted digraph associated with A_L , and note that G and G_{A_L} have the same structure but G_{A_L} has also all the self-loops.

By assumption, G contains a globally reachable node, so also G_{A_L} contains a globally reachable node. Moreover G_{A_L} must be aperiodic since all nodes of G_{A_L} have a self-loop.

Hence, we know that 1 is a simple and strictly dominant eigenvalue of A_L

To conclude, note that $A_L = I - \varepsilon L \Leftrightarrow L = \frac{1}{\varepsilon}(I - A_L)$

$$\Leftrightarrow \lambda \in \text{spec}(A_L), \quad 1/\varepsilon(1 - \lambda) \in \text{spec}(L)$$

This means that 0 is a simple eigenvalue of L

Laplacian matrix and its properties

Q: Are these matrices Laplacian matrices?

(i) $L_{ii} = D_{ii}$ (ii) $L_{ij} \leq 0 \text{ if } i \neq j$ (iii) $L_{ii} \geq 0$

$$L = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 2 & -1 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

✓

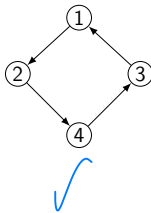
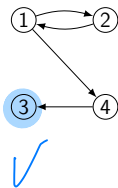
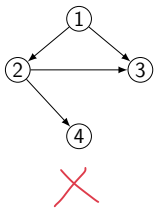
$$L = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

✓

$$L = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

Laplacian matrix and its properties

Q: Do the Laplacians of these digraphs have a simple eigenvalue in 0?



CT linear systems and convergence

Consider a continuous-time, autonomous, linear time-invariant systems:

$$\dot{x}(t) = Ax(t), \quad t \in \mathbb{R}, t \geq 0, \quad x(0) = x_0 \in \mathbb{R}^n$$

whose solution is

$$x(t) = e^{At}x_0, \quad t \in \mathbb{R}, t \geq 0.$$

Def: A matrix $A \in \mathbb{R}^{n \times n}$ is

(i) CT semi-convergent if $\lim_{t \rightarrow +\infty} e^{At}$ exists (finite);

(ii) CT convergent (or Hurwitz stable) if it is semi-convergent and $\lim_{t \rightarrow +\infty} e^{At} = 0_{n \times n}$

How to characterize CT semi-convergent and convergent matrices?

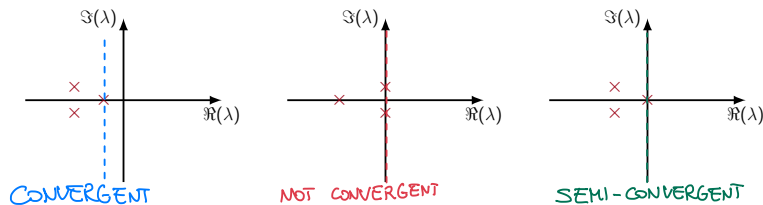
CT linear systems and convergence

Thm (CT convergence and spectral abscissa): Consider a matrix $A \in \mathbb{R}^{n \times n}$ and let $\alpha(A) := \max\{\Re(\lambda) \mid \lambda \in \text{spec}(A)\}$ (spectral abscissa of A). Then:

(i) A CT convergent $\iff \alpha(A) < 0$;

(ii) A CT semi-convergent and not CT convergent $\iff 0$ is a semisimple eigenvalue of A and all the other eigenvalues have real part strictly less than 0.

Q: $A \in \mathbb{R}^{3 \times 3}$ with the following spectrum is CT (semi-)convergent?



Note: The proof follows by bringing A into its J-canonical form:

$$\exists T: A = T J T^{-1} \rightarrow J = \begin{bmatrix} J_1 & & \\ & \ddots & \\ & & J_m \end{bmatrix}$$

$$e^{At} = e^{T J T^{-1} t} = T e^{Jt} T^{-1} = T \begin{bmatrix} e^{J_1 t} & & \\ & \ddots & \\ & & e^{J_m t} \end{bmatrix} T^{-1} \xrightarrow[t \rightarrow \infty]{\lambda \neq 0 \text{ simple}} T \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} T^{-1}$$

CT averaging systems achieving asymptotic consensus

CT averaging system: $\dot{x}(t) = -Lx(t)$, L = Laplacian of weighted digraph G

Thm (Consensus in CT averaging systems): Let G be a weighted digraph with Laplacian matrix L . The following statements are equivalent:

(A1) 0 is a simple eigenvalue of $-L$ and all other eigenvalues μ of $-L$ satisfy $\Re(\mu) < 0$;

(A2) $-L$ is CT semi-convergent and $\lim_{t \rightarrow \infty} e^{-Lt} = \mathbf{1}_n w^\top$ where $w \in \mathbb{R}^n$ satisfies $w \geq 0$, $\mathbf{1}_n^\top w = 1$, and $w^\top L = \mathbf{0}_n^\top$;

(A3) G contains a globally reachable node. *Note: Main diff with DT is that here we do not require aperiodicity*

If any, and therefore all, of the previous conditions are satisfied, then

(i) w is the left dominant eigenvector of $-L$ and $w_i > 0$ iff node i is globally reachable;

(ii) the solution of the CT averaging system satisfies $\lim_{t \rightarrow \infty} x(t) = (w^\top x(0)) \mathbf{1}_n$.

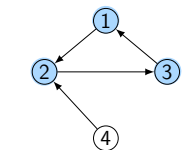
Note: If G is weight-balanced $L \mathbf{1}_n = \mathbf{0}_n \Rightarrow w = \mathbf{1}_n/n$

$$\Rightarrow w^\top x(0) = \frac{1}{n} \sum_{i=1}^n x_i(0) \quad \text{AVG. CONSENSUS}$$

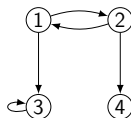
CT averaging systems achieving asymptotic consensus

Q: Do the systems described by the following digraphs achieve asymptotic consensus?

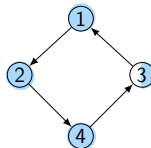
● = globally reachable nodes



CONSENSUS



NO-CONSENSUS



CONSENSUS

Note that in DT there would not be consensus

CT averaging systems: Rosetta stone

